STATISTICAL INSIGHTS FROM STUDENT PERFORMANCE DATASET

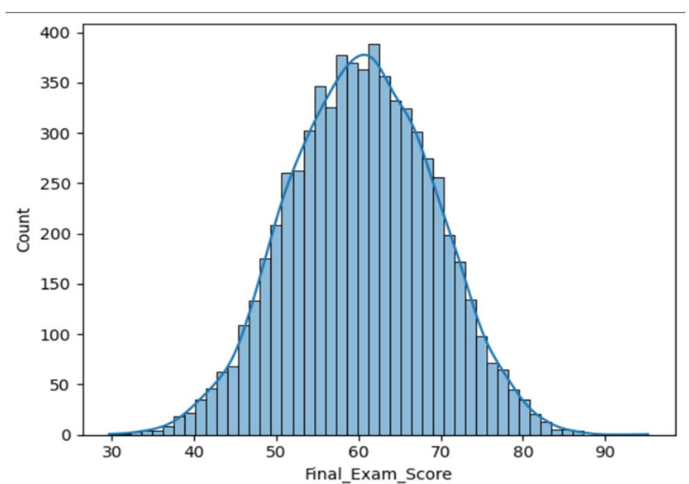
OVERVIEW

This project analyzes student exam performance using statistical techniques implemented in Python. Libraries such as pandas, NumPy, Matplotlib, Seaborn, and SciPy were used for data cleaning, analysis, visualization, and hypothesis testing. Statistical methods including measures of central tendency, correlation analysis, and grouped comparisons were applied to evaluate factors like study hours and attendance. Hypothesis testing was conducted to validate key relationships and support data-driven conclusions.

RESULTS AND INSIGHTS

Distribution of Exam Scores

The distribution of exam scores shows that the mode is 60, identified from both the histogram peak and KDE curve. The mean score is 60.4, which is very close to the mode. Although the graph is slightly right-skewed, the difference is minimal, indicating that the distribution is approximately symmetric. This suggests that most student's scores are concentrated around 60, reflecting a strong central tendency near the average.

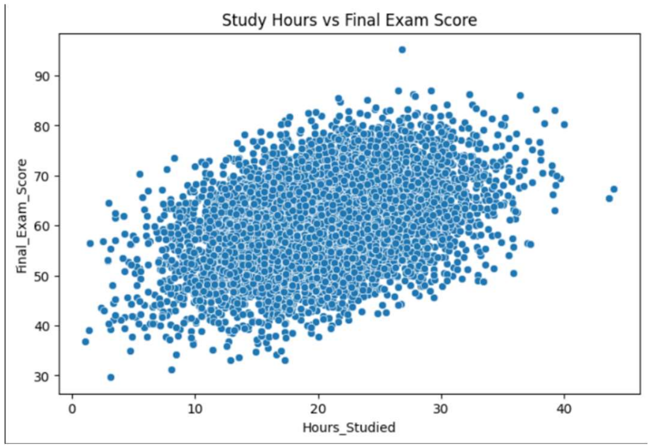


The scatter plot shows a clear upward trend, indicating that final scores tend to increase as study hours increase.

Correlation between Study Hours and Final Marks

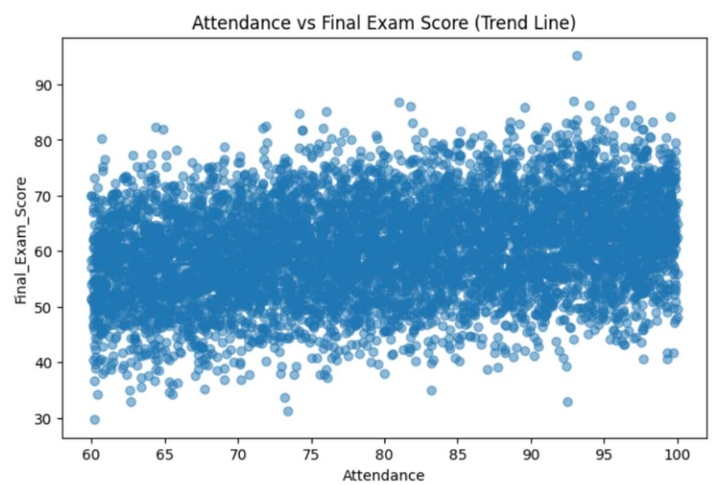
The correlation between study hours and final marks is +0.40985, indicating a moderate positive relationship. This means that as study hours increase, exam scores tend to increase.

However, since the correlation is moderate, it implies that other factors also influence student performance.



Correlation between Attendance and Final Marks

The correlation between attendance and final marks is +0.266, indicating a weak positive relationship. This suggests that higher attendance slightly contributes to better performance, but its effect is limited. Compared to study hours, attendance is a less influential factor in determining exam scores.



The trendline shows a positive relationship between attendance and scores, but it is weaker compared to study hours.

Impact of Parental Involvement

The mean exam scores across parental involvement levels are:

High: 60.25 | Medium: 60.44 | Low: 60.49

The differences are minimal, indicating that parental involvement has no significant impact on student performance in this dataset.

Impact of Internet Access and Learning Resources

Internet Access – Mean Score

No - 60.29 | Yes - 60.41

There is no significant difference in scores, indicating that internet access alone does not influence performance.

Access to Learning Resources – Mean Score

High - 62.26 | Medium - 60.10 | Low: 58.30

Students with higher access to learning resources perform better, indicating that learning resources have a meaningful impact on academic performance.

Student Improvement Analysis

A total of 216 students (approximately 3.2%) improved their scores by at least 5 marks. This indicates that only a small proportion of students showed measurable improvement.

Comparison: Improved vs Not Improved Students

Metric (Mean)	Improved	Not Improved
Study Hours	26.13	20.28
Attendance	86.74	79.75
Previous Scores	55.62	75.38
Sleep Hours	7.10	7.19
Final Exam Score	64.03	60.28

Students who improved generally had higher study hours and better attendance. Additionally, students with lower previous scores showed more improvement, indicating a ceiling effect where high-performing students have less room for growth. Sleep hours showed negligible differences, suggesting minimal impact.

Study Hours Vs Final Score – Does scores stop increasing after a certain number of study hours

Exam scores increase steadily with higher study hours, showing a strong and consistent positive relationship.

There is no evidence that increasing study hours stops being effective within the observed range.

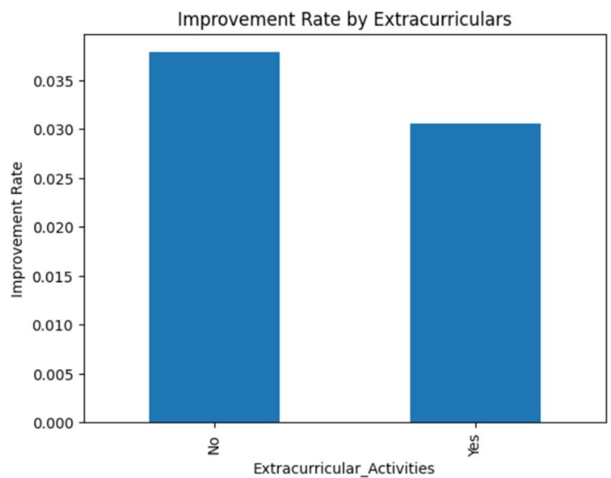
Extracurricular Activities and Improvement

Improvement rates:

With extracurricular activities - 3.1%

Without extracurricular activities - 3.8%

The difference is minimal, indicating that extracurricular participation does not significantly affect academic improvement.



Motivation vs Study Hours (ANOVA Test)

Hypotheses

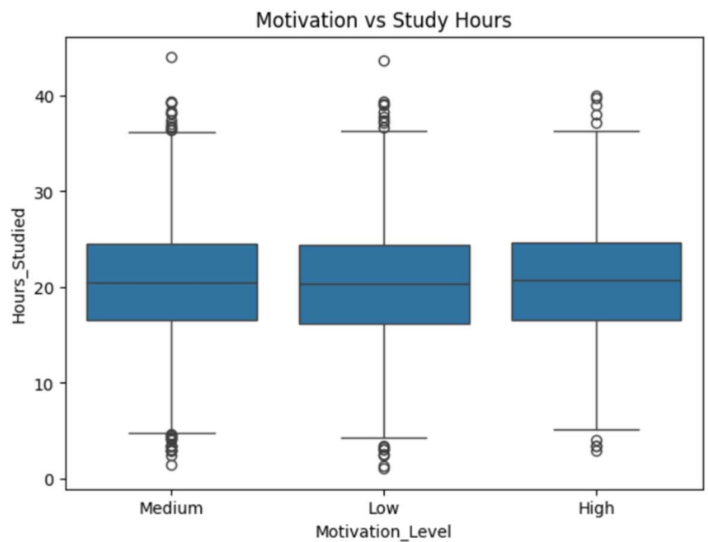
H_0: No difference in study hours across motivation levels

H_1: At least one group differs

We got : F-value = 0.556 | p-value = 0.547 (> 0.05)

Since $p > 0.05$ - we fail to reject the null hypothesis. There is no statistically significant difference in study hours across motivation levels .

Box plot analysis also shows similar median study hours across all groups, with high overlap, confirming that motivation level does not significantly influence study time



Conclusion

- Study hours are the strongest predictor of student performance
- Attendance has a weaker positive impact
- Access to learning resources significantly improves scores
- Factors such as parental involvement, internet access, extracurricular activities, and motivation show minimal or no significant impact in this dataset